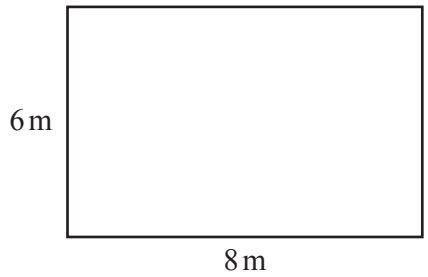
0580_s20_qp_33_Q3

3 **(a)**



NOT TO SCALE

The diagram shows a rectangular patio with sides 6 m and 8 m.

(i) Work out the perimeter of the patio.

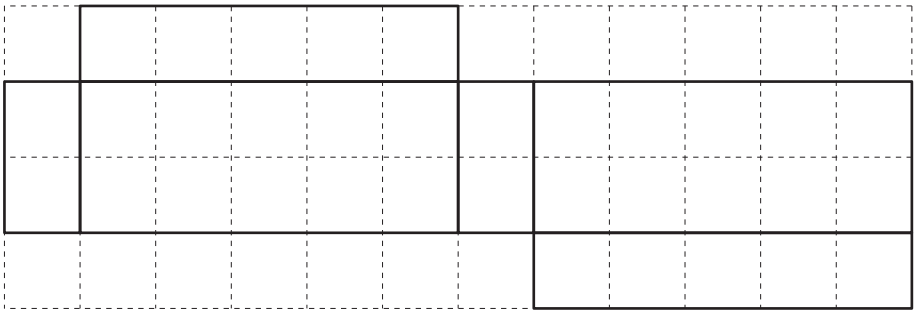
..... m [1]

(ii) Henri covers the patio floor with square tiles.
The tiles are 0.5 m by 0.5 m.

Work out the number of tiles he needs.

..... [2]

(b) The diagram shows the net of a solid on a 1 cm^2 grid.



(i) Write down the mathematical name for the solid.

..... [1]

(ii) Work out the volume of the solid.

..... cm^3 [2]

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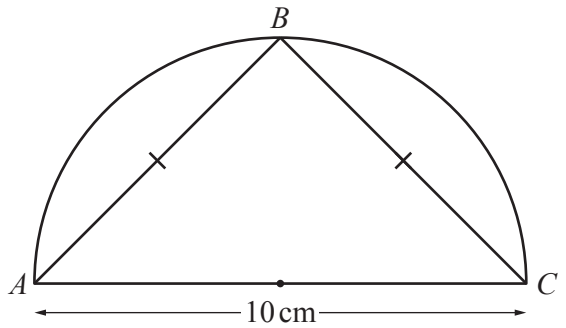
(c) A square has perimeter $12x$.

Find an expression, in terms of x, for the area of the square.
Give your answer in its simplest form.

..... [3]

(d)



NOT TO SCALE

The diagram shows a semicircle with diameter AC.
B is a point on the circumference and $AB = BC$.

Work out the area of triangle ABC.

..... cm^2 [3]